UE24CS352A Machine Learning - Mini-Project Write-up

Reproducing and extending *Quantitative Trading with Machine Learning* (R. M. Glawion, Stanford CS229) | Team: Zia (PES2UG24CS620), Harshavardhan (PES2UG25CS817)

1. Problem statement

Predict the h-day-ahead return of Volkswagen (VW) stock from the lagged daily returns of VW's supply chain plus macro variables, and turn the forecasts into a long/short trading rule. **Part 1** reproduces the paper's four models (Elastic Net, Decision Tree, XGBoost, LightGBM) and its Table 1 (RMSE) and Table 2 (trading profit). **Part 2** adds one model the paper does not use and compares it on the same data, split and metrics.

2. Dataset

We use the author's public replication data (github.com/rglawion/cs229_project_report, Yahoo Finance prices from Frankfurt): daily closes of VW and 30 listed suppliers, 3 Jan 2005 - 30 Dec 2019 (3,809 trading days), plus 13 macro series (Euribor, EUR swap, iTraxx Crossover and their 1/6-month relative changes, output gap, CPI, Bund yield). Features are 27 daily-return lags for each of the 31 stocks + 13 macro variables = **850 features**; missing values are set to 0 as in the paper. Target: VW return over the next h trading days, h = 1, 5, 20 (the horizons shown on the paper's poster).

3. Approach

Validation scheme (as in the paper). Rolling, strictly sequential: 5 years training, 1 year validation (hyper-parameter selection by validation MSE), 1 month out-of-sample test; the window then rolls forward one month. This gives **109 refits** and 2,156 test days (Jan 2011 - Dec 2019) per model and horizon, identical in size to the author's saved results. Metric: RMSE (and MAE), averaged over the monthly test windows as in the author's code.

Trading rule. Start with EUR 1,000; hold one long share if the forecast exceeds threshold c, one short share if it is below -c, no transaction costs (author's rule, ported and verified, see section 5).

New model: RBF Support Vector Regression (SVR). Chosen because (i) it is not used or mentioned in the reference, (ii) the data are small (about 1,300-3,000 rows) and high-dimensional (850 features), where kernel methods are a natural fit, and (iii) it differs from the paper's models: Elastic Net is linear, trees/boosting are axis-aligned, SVR is a smooth non-linear regulariser. Inputs are standardised on the training window; grid: C in {0.1, 1, 10}, epsilon in {0.001, 0.01}, tuned on the validation year every 6 months.

4. Implementation overview

Python 3 (pandas, scikit-learn, xgboost, lightgbm). Everything is in one notebook, `mini_project.ipynb`: data and the 850 features, the rolling 5y/1y/1m windows (109 refits), the five models with grid search, the RMSE/MAE tables, the trading simulator (validated against the paper's Table 2) and the figures. By default it loads the saved test predictions in `results/` and runs in about a minute, including a one-window live demo; `RUN_FULL = True` retrains everything. **Deviations (compute-limited, one CPU instead of the author's 64 cores / ~36 h):** smaller hyper-parameter grids; XGBoost and LightGBM are re-tuned every 12 / 6 months (refit every month); LightGBM uses colsample 0.5 and 63 histogram bins, XGBoost 64 bins; only h = 1, 5, 20 were run. Timings are therefore not comparable with the paper's.

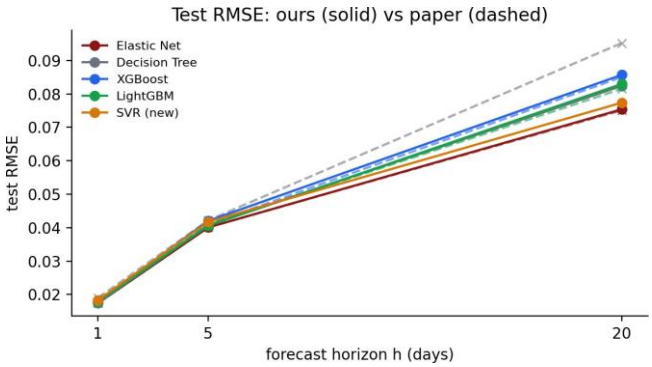
5. Results

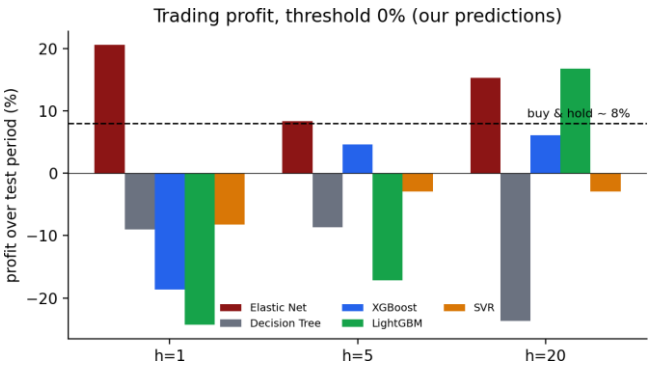
Table A - Prediction error (test RMSE), paper vs. ours. Elastic Net is reproduced almost exactly (within 0.4%); XGBoost and LightGBM within 2.5%. Our Decision Tree is 3-14% *better* than the paper's, most likely because our reduced grid only offers max_features of sqrt/log2 and depth up to 20 (more regularised trees than the paper's full grid); we did not test this explanation.

Model	h	Paper RMSE	Our RMSE	diff	Our MAE
Elastic Net	1	0.0174	0.0174	+0.1%	0.0137
Elastic Net	5	0.0401	0.0401	-0.1%	0.0327
Elastic Net	20	0.0751	0.0754	+0.4%	0.0668
Decision Tree	1	0.0190	0.0175	-8.0%	0.0138
Decision Tree	5	0.0422	0.0407	-3.5%	0.0326
Decision Tree	20	0.0952	0.0824	-13.5%	0.0689
XGBoost	1	0.0176	0.0179	+1.9%	0.0141
XGBoost	5	0.0410	0.0420	+2.5%	0.0346
XGBoost	20	0.0850	0.0857	+0.8%	0.0754
LightGBM	1	0.0177	0.0179	+0.9%	0.0141
LightGBM	5	0.0408	0.0406	-0.6%	0.0333
LightGBM	20	0.0815	0.0830	+1.8%	0.0731
SVR (new)	1	-	0.0183	-	0.0145
SVR (new)	5	-	0.0417	-	0.0342
SVR (new)	20	-	0.0774	-	0.0688

Model	h	Paper %	Our %	Our trades
Elastic Net	1	-9.07	+20.58	111
Elastic Net	5	+9.56	+8.37	229
Elastic Net	20	+21.49	+15.28	75
Decision Tree	1	-8.02	-8.98	137
Decision Tree	5	-3.56	-8.64	97
Decision Tree	20	+7.97	-23.69	441
XGBoost	1	+0.63	-18.67	1251
XGBoost	5	-8.83	+4.61	655
XGBoost	20	+6.57	+6.10	243
LightGBM	1	+9.85	-24.27	1307
LightGBM	5	+5.39	-17.20	859
LightGBM	20	+2.91	+16.81	579
SVR	1	-	-8.18	1429
SVR	5	-	-2.94	601
SVR	20	-	-2.91	427

Left: Table A. Right: Table B - trading profit at threshold 0% over the test period (paper's buy-and-hold: 7.98%; ours 7.8-8.0%).





Trading results are much less reproducible than the RMSEs. We ported the author's simulator and verified it: fed with his saved predictions it matches all 96 cells of the paper's Table 2 exactly (profit and number of trades). However, his code orders predictions within each month by an unstable sort, so the published profit depends on that arbitrary ordering. Re-running with chronological order and our own predictions, the paper's headline cell (Elastic Net, h = 20, c = 0) gives **+15.3%** vs +21.5% published (+20.0% from the author's own predictions in chronological order), i.e. about 1.9-2.5x buy-and-hold rather than 3x; the h = 20 result holds across orderings and across prediction sets, whereas h = 1 and h = 5 profits change sign (e.g. Elastic Net h = 1: +20.6% ours vs -9.1% paper) and should be read as noise. Profits ignore transaction costs and risk, so the large trade counts of boosting/SVR models (400-1,400) make them look better than they would in practice.

New model (SVR) vs. reference models. SVR is **worse than Elastic Net** at every horizon (RMSE 0.0183 vs 0.0174 at h = 1, 0.0417 vs 0.0401 at h = 5, 0.0774 vs 0.0754 at h = 20; +5%, +4%, +3%), **similar to** XGBoost/LightGBM at h = 1 and 5, and **better than** Decision Tree, XGBoost and LightGBM at h = 20 (0.0774 vs 0.0824 / 0.0857 / 0.0830), where it is second only to Elastic Net. Its validation-selected C was mostly the smallest (0.1), i.e. a strongly regularised, nearly smooth fit. Our interpretation (not tested by ablation): daily-return features carry a weak signal buried in noise, so heavily regularised, smooth models win and flexible tree ensembles overfit; SVR benefits from regularisation but lacks Elastic Net's feature selection across 850 correlated inputs.

6. Conclusions

(1) The paper's prediction results are reproducible: Elastic Net is the most accurate model at h = 1, 5 and 20, and RMSE grows with the horizon. (2) The added RBF-SVR is competitive but does not beat Elastic Net; it ranks second at h = 20. (3) The paper's trading claims should be treated cautiously: they depend on the ordering of predictions inside each month, short horizons are unstable, and only the 20-day horizon shows a consistent edge (about 2x buy-and-hold before costs). **Limitations:** reduced grids and tuning schedule, h = 10/15/25 not run, one random seed (trees/boosting), no transaction costs or risk measures.